PHY102 Electricity and Magnetism Jan-April 2026: Assignment 2

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To be discussed in the Tutorial on Jan 23

1. Two balls 1 kg each are hanging vertically via a nylon(non-extendable) wire from a common suspension point. The length of the string is one meter. The balls are charged and because of the repulsion they are at a distance 10cm apart. Taking $g = 10m/sec^2$ calculate the charge on each ball (assuming that the charge on each ball is the same).

2. Consider a vector field
$$\vec{A}(x, y, z) = \hat{z} + \frac{z}{10}(\hat{x} + \hat{y})$$
Consider a vertical cylinder of radius R and height H, with center of one flat face at the origin.
   (a) Draw the vector field arrows and the cylinder.
   (b) Calculate the flux of the vector field through the two flat faces and the curved surface of the cylinder.

3. Consider water flowing in space with a velocity field described by
$$\vec{v}(x, y, z) = v_0(\hat{x} + \frac{1}{10}\hat{y})$$
. Consider a cube of edge length a, and with its edges along the x, y and z axes. Calculate the flux of the velocity field through all the faces of the cube. What is the total flux? Now imagine that this is not a velocity field but it is an electric field coming from a charge distribution. What can you say about the charge inside the cube?

4. Consider a uniformly charged sphere of radius R. Using Gauss law calculate the field for all $0 \leq r \leq \infty$. What do you observe?

5. **(a)** Consider a charge distribution
$$\rho = \rho_0 \left(\frac{1}{r}\right).$$
Calculate the electric field produced by such a charge distribution by using Gauss law in all space $0 \leq r \leq \infty$.
   (b) Now consider a charge distribution:
$$\rho = \rho_1 \left(\frac{1}{r}\right) + \rho_2 \left(\frac{1}{r - r_0}\right)$$
where r_0 is a constant. Is the above distribution spherically symmetric? Can you calculate the electric field generated by this charge distribution? Hint the solution of part (a) may be a useful motif.

6. Consider two infinite plates with uniform surface charge density σ in the $x - y$ plane, one at $z = -a/2$ and the second at $z = +a/2$. Using Gauss law compute the electric field in entire space. Repeat the calculation when the plate at $z = +a/2$ has uniform charge density $-\sigma$ instead of σ. In both cases, calculate the energy density associated with the electric field as a function of σ.